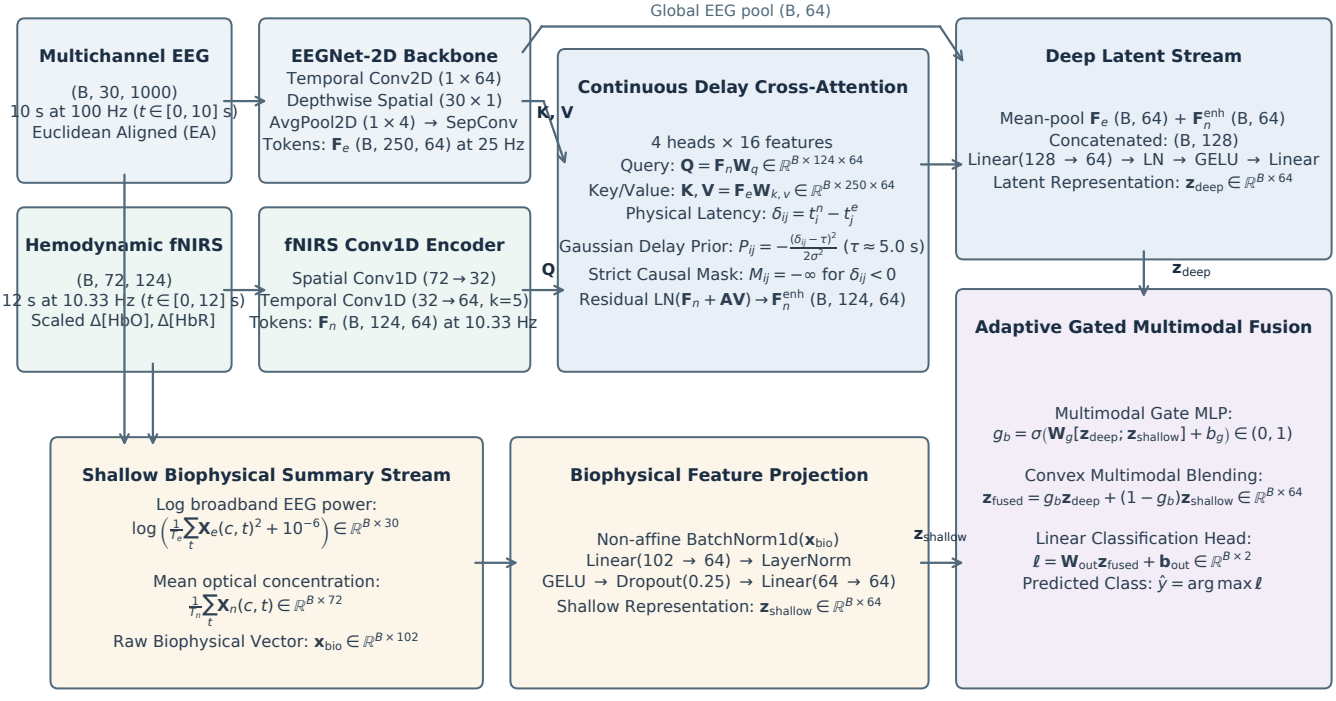

Global EEG pool (B, 64)
Multichannel EEG
(B, 30, 1000)
10 s at 100 Hz ($t \in [0, 10]$ s)
Euclidean Aligned (EA)
EEGNet-2D Backbone
Temporal Conv2D (1×64)
Depthwise Spatial (30×1)
AvgPool2D (1×4) → SepConv
Tokens: $\mathbf{F}_e$ (B, 250, 64) at 25 Hz
K, V
Continuous Delay Cross-Attention
4 heads × 16 features
Query: $\mathbf{Q} = \mathbf{F}_n \mathbf{W}_q \in \mathbb{R}^{B \times 124 \times 64}$
Key/Value: $\mathbf{K}, \mathbf{V} = \mathbf{F}_e \mathbf{W}_{k,v} \in \mathbb{R}^{B \times 250 \times 64}$
Physical Latency: $\delta_{ij} = t_i^n - t_j^e$
Gaussian Delay Prior: $P_{ij} = -\frac{(\delta_{ij} - \tau)^2}{2\sigma^2}$ ($\tau \approx 5.0$ s)
Strict Causal Mask: $M_{ij} = -\infty$ for $\delta_{ij} < 0$
Residual LN($\mathbf{F}_n + \mathbf{AV}$) → $\mathbf{F}_n^{\text{enh}}$ (B, 124, 64)
Deep Latent Stream
Mean-pool $\mathbf{F}_e$ (B, 64) + $\mathbf{F}_n^{\text{enh}}$ (B, 64)
Concatenated: (B, 128)
Linear(128 → 64) → LN → GELU → Linear
Latent Representation: $\mathbf{z}_{\text{deep}} \in \mathbb{R}^{B \times 64}$
Hemodynamic fNIRS
(B, 72, 124)
12 s at 10.33 Hz ($t \in [0, 12]$ s)
Scaled Δ[HbO], Δ[HbR]
fNIRS Conv1D Encoder
Spatial Conv1D (72 → 32)
Temporal Conv1D (32 → 64, k=5)
Tokens: $\mathbf{F}_n$ (B, 124, 64) at 10.33 Hz
Q
$\mathbf{z}_{\text{deep}}$
Adaptive Gated Multimodal Fusion
Multimodal Gate MLP:
$g_b = \sigma(\mathbf{W}_g[\mathbf{z}_{\text{deep}}; \mathbf{z}_{\text{shallow}}] + b_g) \in (0, 1)$
Convex Multimodal Blending:
$\mathbf{z}_{\text{fused}} = g_b \mathbf{z}_{\text{deep}} + (1 - g_b)\mathbf{z}_{\text{shallow}} \in \mathbb{R}^{B \times 64}$
Linear Classification Head:
$\ell = \mathbf{W}_{\text{out}} \mathbf{z}_{\text{fused}} + \mathbf{b}_{\text{out}} \in \mathbb{R}^{B \times 2}$
Predicted Class: $\hat{y} = \arg\max \ell$
Shallow Biophysical Summary Stream
Log broadband EEG power:
$\log\left(\frac{1}{T_e}\sum_t \mathbf{X}_e(c,t)^2 + 10^{-6}\right) \in \mathbb{R}^{B \times 30}$
Mean optical concentration:
$\frac{1}{T_n}\sum_t \mathbf{X}_n(c,t) \in \mathbb{R}^{B \times 72}$
Raw Biophysical Vector: $\mathbf{x}_{\text{bio}} \in \mathbb{R}^{B \times 102}$
Biophysical Feature Projection
Non-affine BatchNorm1d($\mathbf{x}_{\text{bio}}$)
Linear(102 → 64) → LayerNorm
GELU → Dropout(0.25) → Linear(64 → 64)
Shallow Representation: $\mathbf{z}_{\text{shallow}} \in \mathbb{R}^{B \times 64}$
$\mathbf{z}_{\text{shallow}}$